

AI Assessment System — current state

Everything below was pulled directly from the repository on disk, the local dev database, and the installed test suite just now — not recalled from prior conversation. Section 6 is explicit about which database it could and couldn't reach.

340 tests passing**0** uncommitted files**10** tables**22** API routes**local DB empty** 0 curricula on disk**10** open items (§9)

01 Schema — actual, not spec'd

Pulled directly from `app/models/*.py`. Columns with a dotted underline were added after their table already existed in production and only exist today because of the self-healing `PENDING_COLUMNS` mechanism in `app/db/schema_sync.py` — `create_all()` alone would never have added them to a live table.

curricula

the central row — standalone AND curriculum-upload entries share this table

COLUMN	TYPE	NOTES
id	String(36) PK	uuid4
topic	Text	not null
target_completion_date	Date	not null — drives all scheduling math
extracted_content	Text	nullable
mastery_achieved	Boolean	nullable
completed_at	DateTime	nullable
status	Enum CurriculumStatus	pending / analyzing / ready / complete
priority	Integer	nullable — reserved, unused
is_active_focus	Boolean	not null, default false — unused
created_at	DateTime	not null
entry_type	Enum CurriculumEntryType	nullable — null = standalone; self-healed
upload_id	String(36) FK curriculum_uploads.id	nullable; self-healed
chapter_label	Text	display only, never parsed; self-healed
max_marks	Float	nullable; self-healed
resources_hold	Boolean	not null default false; self-healed
last_hold_reminder_at	DateTime	nullable; self-healed

assessments

COLUMN	TYPE
id	String(36) PK
curriculum_id	FK curricula.id
attempt_number	Integer, default 1
assessment_text / rubric	Text, nullable
<u>part1_text / part1_rubric</u>	Text, nullable
<u>part2_text / part2_rubric</u>	Text, nullable
duration_minutes	Integer, nullable
generation_prompt_id	FK prompt_templates.id
scheduled_at / reminder_at / due_date	DateTime, not null
status	Enum AssessmentStatus
submission_token	String(255) unique
scheduled_job_ids	JSON
<u>send_job_claimed_at</u>	DateTime – TOCTOU guard
created_at	DateTime

grades

COLUMN	TYPE
id	String(36) PK
submission_id	FK submissions.id, unique
mastery_score	Float, not null
weak_areas	JSON
overall_feedback	Text, not null
<u>part1_score / part2_score</u>	Float, nullable
<u>score_earned</u>	Float – GPA numerator input
<u>max_marks</u>	Float – denormalized snapshot
grading_prompt_id	FK prompt_templates.id
graded_at	DateTime

curriculum_uploads

COLUMN	TYPE
id	String(36) PK
source_filename	Text, not null
note_on_pending_resources	Text, nullable
chapters_with_no_standalone_assessment	JSON, nullable
uploaded_at	DateTime, not null
syllabus_email_sent_at	DateTime, nullable
course_material_snapshot / captured_at	JSON / DateTime
<u>closed_at</u>	DateTime – soft- archive marker
<u>last_secondary_transcript_sent_at</u>	DateTime – added this session, see §9

midterm_details 1:1 with curricula

COLUMN	TYPE
id	String(36) PK
curriculum_id	FK curricula.id, unique
known_now	JSON list, not null
pending_completion_labels	JSON dict {slug: label}
pending_completion_slots	JSON dict {slug: value null}
probe_focus / special_case	Text, nullable
part1_max_marks / part2_max_marks	Float, not null

submissions

COLUMN	TYPE
id	String(36) PK
assessment_id	FK assessments.id, unique
submission_type	Enum: github_url / text / file
github_url / text_content / file_path	Text, nullable – exactly one set
<u>part1_text_content</u>	Text – midterm Part 1 answer
submitted_at	DateTime
raw_payload	JSON – audit copy of the POST body

late_submission_tokens

COLUMN	TYPE
id	String(36) PK
issued_at	DateTime
used_at / used_by_assessment_id	nullable – spent marker
<u>curriculum_upload_id</u>	FK, nullable – null = standalone pool

resources

id, curriculum_id FK, type
(url/pdf/note/github_repo/markdown), raw_content,
source_ref. Unmodified since the original build.

reschedule_requests

id, assessment_id FK, reason_text, classification_category,
category_reasoning, approved, requested_at. Unmodified.

prompt_templates

id, slug, version, body, is_active, created_at. Unique on
(slug, version). 6 valid slugs incl.

midterm_generation / midterm_grading – see §9 on a
gap in what health-checks for.

02 API surface

22 routes across 8 routers, all mounted under `/api/v1` (health under `/health`). **API-only** marks a route with no corresponding button/action in `app/static/index.html` — confirmed by grepping every `fetch()` call the UI actually makes, not inferred.

Curriculum – standalone

METHOD	PATH	REQUEST	RESPONSE
POST	<code>/curriculum/</code>	<code>form: topic, target_completion_date, links[], github_repos[], notes, files[]</code>	<code>{curriculum_id}</code>
GET	<code>/curriculum/{id}</code>	–	CurriculumResponse API-only

Assessments

METHOD	PATH	REQUEST	RESPONSE
POST	<code>/assessments/{curriculum_id}</code>	–	AssessmentSummary API-only manual create+schedule
GET	<code>/assessments/{id}?token=</code>	<code>query: token</code>	AssessmentDetailResponse – lazily generates content for retroactive-expired entries
POST	<code>/assessments/{id}/reschedule</code>	<code>{token, reason}</code>	<code>{approved, reasoning, new_scheduled_at, new_due_date}</code>

Submissions & grading

METHOD	PATH	REQUEST	RESPONSE
POST	<code>/submissions/</code>	<code>form: assessment_id, token, submission_type, github_url?, text_content?, file?, part1_text_content?</code>	<code>{submission_id}</code>
GET	<code>/submissions/{id}</code>	–	<code>{submission_id}</code>
GET	<code>/submissions/{id}/results</code>	–	GradeResponse

Curriculum uploads

METHOD	PATH	REQUEST	RESPONSE	
POST	/curriculum-uploads/	multipart: file (JSON)	{upload_id, entry_count, syllabus_email_sent}	
GET	/curriculum-uploads/{id}	–	entries[] with status/hold/pending-slots	
GET	/curriculum-uploads/{id}/gpa	–	{total_earned, total_max, gpa, graded_count, missed_count}	API-only
GET	/curriculum-uploads/{id}/transcript	–	chapter_groups[] + totals	API-only
POST	/curriculum-uploads/{id}/entries	{entry}	CurriculumEntrySummary	
PATCH	/curriculum-uploads/entries/{id}	{updates}	CurriculumEntrySummary	
PATCH	/curriculum-uploads/entries/{id}/pending-resources	{values}	{resources_hold, pending_completion_slots}	
POST	/curriculum-uploads/{id}/close	–	{upload_id, closed_at}	

Late tokens

METHOD	PATH	
GET	/late-tokens/? curriculum_upload_id=	
POST	/late-tokens/grant? curriculum_upload_id=	API-only

Health

METHOD	PATH	
GET	/health/prompts	API-only

03 Scheduled jobs

APScheduler 3.x, SQLAlchemyJobStore , single worker (enforced by a comment in `render.yaml` — multiple workers would double-fire everything below).

Restart-vulnerability class: one job in this system used to be `trigger="interval"` (`send_biweekly_transcript` , removed this session). An interval trigger's `next_run_time` is computed as *registration time + N days* — since `start()` re-registers every job on every process boot with `replace_existing=True` , a server that restarts more often than the interval could push that send arbitrarily far into the future. It's been replaced with a persisted-timestamp check inside an existing `cron` job (see below) — cron anchors to an absolute calendar instant, not a relative offset, so restarts don't move it. No other recurring job in the table below carries this risk.

JOB	TRIGGER	CADENCE	READS / WRITES
<code>grant_late_tokens_monthly</code>	<code>cron</code>	<code>day=1, hour=0, minute=0</code> — 1st of every month, midnight	writes: tops up every late-token pool (standalone + one per curriculum_upload) to the 2-token cap
<code>recheck_pending_midterms_daily</code>	<code>cron</code>	<code>hour=6, minute=0</code> — daily	(a) clears resource-holds whose pending slots are now filled, or re-sends the hold reminder past its 7-day throttle; (b) new this session — for every open upload, sends the secondary-recipient transcript copy if <code>transcript_secondary_interval_days</code> real days have elapsed since <code>last_secondary_transcript_sent_at</code> (or <code>uploaded_at</code> , if never sent)
<code>send_reminder_{assessment_id}</code>	<code>date</code>	one-shot, at <code>reminder_at</code>	sends the pre-exam reminder email
<code>assessment_{assessment_id}</code>	<code>date</code>	one-shot, at <code>scheduled_at</code>	claims the row atomically (TOCTOU guard), generates content for upload entries at send-time, sends the exam email, flips status to active
<code>expire_{assessment_id}</code>	<code>date</code>	one-shot, at <code>due_date</code>	flips status active → expired if still unsubmitted
<code>grade_{submission_id}</code>	<code>date</code>	one-shot, at submission time (<code>run_date = now</code>)	grades, marks mastery or schedules a capped retest, sends results + (for uploads) a fresh transcript

All jobs share `misfire_grace_time=None, coalesce=True` — a job missed while the server was down fires once on the next start, rather than being dropped or replayed per missed tick.

AssessmentService untouched this session

- `create_for_curriculum` — first-attempt assessment, branches to midterm generation for `entry_type=midterm`
- `create_retest` / `_create_midterm_retest` — second-attempt, weak-areas-focused
- `generate_assessment_content` / `generate_midterm_content` — deferred to send-time for upload entries
- `_assemble_midterm_pool` — cumulative-Assessment pool for Part 1
- `get_by_id_and_token` — submission-link auth

CurriculumService untouched this session

- `create` — standalone ingestion, LLM curriculum analysis, first assessment, single commit boundary
- `mark_mastery` — wired into `grade_submission_job` this build; was dead code before

GradingService untouched this session

- `grade` / `_grade_midterm` — single-score vs two-part scoring
- `get_results` — read path for the results endpoint

GPAService untouched this session

- `compute_gpa` — pure, recomputed live on every read, never cached; fixed for a stale-retake bug earlier in this build (only the final attempt counts)

TranscriptService untouched this session

- `display_status` — live classifier: Held / Not Yet Due / Missed (Late-Eligible or No Score) / Exam Sent / Graded
- `compute_transcript` — full chapter-grouped assembly, uncached

LateTokenService untouched this session

- `grant_monthly` / `spend` / `get_balance` — each scoped per `curriculum_upload_id` (null = standalone), independent 2-token pools since the parallel-curricula isolation fix

RescheduleService /**SchedulerService** /**SubmissionService** untouched this session

- Reschedule: LLM-classifies the excuse into approved/denied categories, applies but never decides approval itself
- Scheduler: thin wrapper translating job-lifecycle calls onto `SchedulerInterface`
- Submission: token-validated create, spends a late token when past `due_date`

CurriculumUploadService modified this session

the largest, most-changed service in the codebase

- `ingest` / `_parse_entries` — validates and classifies every entry on upload (schedule now, hold, or retroactively expired)
- `add_entry` / `update_entry` / `close_upload` — entry lifecycle + soft-archive, added earlier this build
- `fill_pending_resources` / `check_and_clear_hold` / `send_hold_reminder` — the Midterm resource-hold state machine
- `send_periodic_transcript_if_due` — **new this session**; replaces the old interval-triggered job, gates on a persisted timestamp vs. wall clock instead of scheduler state

EmailService docstring updated this session

6 send methods (assessment, reminder, results, syllabus, transcript, midterm-hold-reminder) — only

`send_transcript_email` 's docstring changed, to point at the new job instead of the deleted one

05 Configuration — current real values

Local `.env` values are things I can read directly. Render dashboard values marked `sync:` `false` in `render.yaml` are **not visible to me** — I have no shell or dashboard access to the live service, so those rows report what's committed, not what's actually configured on Render.

SETTING	LOCAL .ENV	RENDER.YAML	
DATABASE_URL	sqlite:///./assessment.db	sqlite:///data/assessment.db (persistent disk)	
ANTHROPIC_API_KEY	configured	<i>sync: false — dashboard only</i>	unv
RESEND_API_KEY	configured	<i>sync: false — dashboard only</i>	unv
RESEND_FROM_EMAIL	onboarding@resend.dev	<i>sync: false — dashboard only</i>	san
USER_EMAIL	mridula.sureshrao@proton.me	<i>sync: false — dashboard only</i>	
TRANSCRIPT_SECONDARY_RECIPIENT_EMAIL	snehalmuskawad11@gmail.com	<i>sync: false — dashboard only</i>	unv
TRANSCRIPT_SECONDARY_INTERVAL_DAYS	15	15 committed	
APP_BASE_URL	http://localhost:8000 (dev-only, fine)	https://REPLACE-WITH-YOUR-RENDER-URL.onrender.com	sti
SUBMISSION_TOKEN_SECRET	configured	generateValue: true (Render mints its own)	
ASSESSMENT_WINDOW_MIN/MAX_DAYS	1 / 3	— (code default 1/3)	
REMINDER_HOURS_BEFORE	24	— (code default 24)	
MASTERY_THRESHOLD	85.0	85.0	
LLM_MAX_RETRIES	3	3	
LLM_MODEL	not set — see note below	claude-sonnet-4-6	loc
pending_hold_reminder_interval_days	unset → default 7	unset → default 7	
entry_reminder_hours_before_deadline	unset → default 24	unset → default 24	
syllabus / exam / results recipient overrides	all unset	all unset	fal
test_mode	unset → false	unset → false	con

Two things worth your attention here: (1) `RESEND_FROM_EMAIL` is still Resend's sandbox sender (`onboarding@resend.dev`) everywhere I can see it — sandbox Resend accounts typically only deliver to the account owner's own verified address, so mail to the secondary transcript recipient could be silently dropped if this hasn't been changed on a verified custom domain in the dashboard. (2) `render.yaml` 's committed `APP_BASE_URL` is still the literal

placeholder string — if that's what's actually live, every emailed exam/reminder link is broken. Neither is something I can confirm without dashboard access.

Also: local `.env` sets `ANTHROPIC_MODEL`, but the settings field is `llm_model`, read from env var `LLM_MODEL` — that line has zero effect locally. Currently harmless because the ignored value and the code default happen to be identical.

06 Real current data state

I can only reach the local dev database (`./assessment.db`) from this session — I have no shell or disk access to Render's production SQLite file. Everything below is the local file, read-only, queried just now with `sqlite3` . It is *not* the deployed data the earlier conversation discussed (PM System, JS Internals, etc.) — that only exists on Render's persistent disk.

Row counts – local assessment.db, queried just now

TABLE	ROWS
curricula	0
curriculum_uploads	0
assessments	0
submissions	0
grades	0
resources	0
midterm_details	0
reschedule_requests	0
late_submission_tokens	2 (unused, standalone pool – curriculum_upload_id is null on both)
prompt_templates	8 (seeded, matches SEED_TEMPLATES)

There is no uploaded curriculum or entry data in the local database at all — so the "is Missed/Late-Eligible sitting correctly" check from the request can't be answered against real entry data from here. Confirming that requires hitting the deployed service directly (e.g. `GET ../curriculum-uploads/{id}/transcript` on the live Render URL) or pulling the production disk file — both out of scope for this read-only, no-live-calls pass.

One live, concrete finding while querying: `SELECT last_secondary_transcript_sent_at FROM curriculum_uploads` failed with `no such column` against this exact local file — direct, current proof of the §1 schema-drift class, for the column added this session. It will self-heal automatically the next time the local app process starts (lifespan → `sync_schema`); I did not run the app or write to this file during this pass.

340 passed

0 failed

0 skipped / xfail

17 files

Coverage by category

- Standalone flow end-to-end — exam scheduling, submission, grading, reschedule
- Curriculum-upload lifecycle — ingestion/classification, modify/close, entry lifecycle integration, parallel-curricula isolation
- GPA and transcript computation, including the periodic secondary-recipient send and its restart-safety (new this session)
- Anthropic adapter contract — retry logic, error mapping, schema validation
- Email flow content and recipient routing
- Real (but isolated, tmp-DB) APScheduler firing semantics
- DB-isolation guard itself, and the schema self-heal mechanism

No "live-only" test category exists — there's no pytest marker for it because every test is fake-adapter-only by construction. Confirmed directly: every `anthropic.*` / `resend.*` / `httpx` reference across the four files that mention them is either a mocked exception class or a monkeypatched send function — never a real network call. The default run makes zero real API calls.

One latent (not currently exploited) gap found while checking this: the shared `client` fixture in `tests/conftest.py` overrides `get_db`, `get_scheduler_service`, `get_assessment_service`, `get_grading_service`, and `get_reschedule_service` with fakes — but not `get_curriculum_service` or `get_curriculum_upload_service`. No test today calls those two routes through the shared client, so the current run is safe. But either dependency closes over the real module-level `_llm` / `_email` singletons (built from real `.env` keys at import time) rather than being overridden — the first test written against `POST /curriculum/` or `POST /curriculum-uploads/` using the plain `client` fixture would silently fire a real Anthropic and/or Resend call.

working tree **clean** — 0 uncommitted files, checked just now

vs. origin/main **in sync** — 0 ahead, 0 behind

this conversation's work fully committed and pushed — nothing discussed here is sitting uncommitted

Recent commits

60aa8a4	Adding secondary email and interval days fix	this session
68d87a7	Fix for 500 Internal Server Error - v1/curriculum-uploads/	
eea8b7f	Fix generation-side README mislabeling and LLMValidationError tool-budget exhaustion	
bd7c69b	Fix GPA stale-retake bug, add midterm README spot-check grading, and build the pending-resources UI	
7154fe2	Add parallel-curricula isolation, entry modify/close, retest-crash fix, isolated test mode, and curriculum management UI	
e30eddb	Adding curriculum upload, midterms, retake cap, GPA, and transcript	original feature build
175d420	Generating tokens	
d08658c	Adding late submission tokens - 2 per month (non-stackable)	

09 Known-open list

From direct working knowledge of this build, verified against the code just now rather than recalled — not waiting to be asked about any one of these.

01 RESEND_FROM_EMAIL is still the Resend sandbox default.

Every place I can see it, it's `onboarding@resend.dev`. Sandbox Resend accounts generally only deliver to the account's own verified address — the secondary-recipient transcript send could be silently no-op'ing or bouncing right now, and I have no way to confirm either way without dashboard access.

02 render.yaml's committed APP_BASE_URL is still the literal placeholder.

`https://REPLACE-WITH-YOUR-RENDER-URL.onrender.com`. If a Blueprint sync ever re-applies this file's `value:` keys, every assessment/reminder email link breaks. The app's own startup warning only checks for `"localhost"` in the URL — it would not catch this specific placeholder.

03 Render dashboard `sync:false` values are unverifiable from here.

`ANTHROPIC_API_KEY`, `RESEND_API_KEY`, `RESEND_FROM_EMAIL`, `USER_EMAIL`, `TRANSCRIPT_SECONDARY_RECIPIENT_EMAIL` — whether this session's `TRANSCRIPT_SECONDARY_*` work has actually taken effect on the live service has never been confirmed by me hitting it; I have no shell or dashboard access.

04 Test client fixture doesn't override the two riskiest LLM-touching routes.

`get_curriculum_service` and `get_curriculum_upload_service` aren't in `tests/conftest.py`'s override list — see §7. Not exploited today, but a real footgun for the next person adding an HTTP-level test there.

05 No HTTP-layer test for the two ingestion routes.

`POST /curriculum/` and `POST /curriculum-uploads/` are only exercised at the service layer with fake adapters directly — route-level multipart/JSON-file parsing for these two specific endpoints is comparatively less covered than everything downstream of them.

06 Per-flow recipient overrides exist in code but aren't configured anywhere.

`syllabus_recipient_email`, `exam_recipient_email`, `results_recipient_emails_raw` are all unset locally and on Render — every one of the 5 curriculum-upload email flows currently falls back to the single `user_email`, even though the config was built to let them diverge.

07 **.env's ANTHROPIC_MODEL line does nothing.**

The settings field is `llm_model`, mapped from env var `LLM_MODEL`, not `ANTHROPIC_MODEL`. Harmless today only because the ignored value and the code default are identical — dead, misleading config.

08 **Local dev DB is schema-drifted right now.**

Missing `curriculum_uploads.last_secondary_transcript_sent_at`, confirmed by a live query failure this pass (§6). Self-heals automatically on the next local app start; not written to during this audit.

09 **Zero visibility into Render's production disk from this session.**

Everything in §6 is the local dev DB only, which currently holds no curriculum/assessment data. Whether real production entries are sitting in the correct Missed/Late-Eligible state can only be confirmed by querying the live service directly, which is out of scope for a read-only, no-side-effect pass.

10 **REQUIRED_SLUGS doesn't include the two midterm prompt templates.**

`/health/prompts` only checks `assessment_generation`, `curriculum_analysis`, `retest_generation`, `grading`, `reschedule_classification`. If `midterm_generation` or `midterm_grading` were ever missing/deactivated, health would still report "ok" while every midterm-type entry failed at generation or grading time.

Generated by direct inspection of the repository, local DB, and test run — not from memory of prior conversation.

All 9 sections read-only · no writes · no real LLM/email calls triggered